

Linkage Disequilibrium

May 13, 2026

1 Phase is unknown

When scoring genotypes (e.g. microsatellites), the phase is unknown meaning that we don't know whether alleles from two different loci are on the same or different chromosomes. For example, for a heterozygous individual $A_1A_2B_1B_2$, A_1 and B_1 can be on the same chromosome (A_1B_1/A_2B_2) or different chromosomes (A_1B_2/A_2B_1).

Because of this issue, a composite measure of linkage disequilibrium is defined:

$$\Delta_{AB} = p_{AB} + p_{A/B} - 2p_Ap_B$$

p_{AB} is the proportion of gametes with two alleles at the same chromosome. $p_{A/B}$ is the proportion of gametes with the two alleles at different chromosomes.

This approach does not require genotype proportions to be in Hardy-Weinberg equilibrium.

A method for more than two alleles is described in [1]. For two loci, A and B, locus A has J alleles and locus B has K alleles. The composite measure of linkage disequilibrium is defined as:

$$\Delta_{A_jB_k} = P(A_jB_k \text{ on the same or different haplotypes}) - 2p_{A_j}p_{B_k}$$

$P(A_jB_k)$ is the observed proportion of gametes with alleles A_j and B_k on the same or different haplotypes. p_{A_j} and p_{B_k} are the observed allele frequencies

of A_j and B_k .

An estimate of $\Delta_{A_j B_k}$ can be estimated is:

$$\hat{\Delta}_{A_j B_k} = \frac{n_{A_j B_k}}{N} - 2\hat{p}_{A_j}\hat{p}_{B_k}$$

$n_{A_j B_k}$ is the number of gametes with alleles A_j and B_k on the same or different haplotypes. N is the total number of gametes. $\hat{p}_{A_j}$ and $\hat{p}_{B_k}$ are the observed allele frequencies of A_j and B_k .

Are there $J \times K$ or $(J - 1) \times (K - 1)$ number of composite coefficients?

A composite correlations for a pair of alleles is defined by [2]:

$$r_{A_j B_k}^2 = \frac{\Delta_{A_j B_k}^2}{[p_{A_j}(1 - p_{A_j}) + D_{A_j}][p_{B_k}(1 - p_{B_k}) + D_{B_k}]}$$

D_{A_j} and D_{B_k} are the Hardy-Weinberg disequilibrium coefficients: $D = \text{observed homozygote proportion} - \text{expected homozygote proportion} (p^2)$.

A correlation-based statistic T_2 is proposed by [2]:

$$T_2 = \frac{(J - 1)(K - 1)N}{J \cdot K} \sum_{j=1}^J \sum_{k=1}^K r_{A_j B_k}^2$$

N is the total number of individuals. [How to deal with missing genotype at some loci?](#)

T_2 follows a χ^2 distribution:

$$T_2 \sim \chi_{(J-1)(K-1)}^2$$

References

- [1] Schaid DJ. 2004. Linkage disequilibrium testing when linkage phase is unknown. *Genetics*. 166(1):505-12. doi: 10.1534/genetics.
- [2] Zaykin DV, Pudovkin A, Weir BS. 2008. Correlation-based inference for linkage disequilibrium with multiple alleles. *Genetics*. 180(1):533-45. doi: 10.1534/genetics.108.089409.